Highlights

Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

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- A single 5 ns MD label has reliability 0.47, capping model correlation near 0.69
- Initialization bias: the first 5 ns of a run are 1.6 times faster than later ones
- About half of the run-to-run noise comes from the initial structure
- Replicating only candidates near the cut-off halves the cost of finding the top 10%
- Lithium and anion diffusivities and ionic conductivity give the same conclusions

Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

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Abstract

High-throughput molecular dynamics (MD) supplies the labels on which materials screening and machine-learning models are built, yet each label comes from one short, noisy run whose value can depend on the simulated time. Treating these labels as stochastic simulation output, we quantify their bias and noise, and how a simulation budget should be spent given both, using HTP-MD, the largest public set of polymer electrolyte trajectories (6,137 runs of 5 ns and 133 of 50 ns). Only 2 short and 5 long runs meet an operational criterion for diffusive lithium motion, and the first 5 ns of a run are 1.63 times faster than its second half while the polymer is still relaxing, an initialization bias that inflates short labels. From 94 randomly sampled systems simulated twice, a single 5 ns label has reliability 0.47 (95% CI 0.31 to 0.62): no model can correlate with such labels much beyond about 0.69, a random forest on monomer fingerprints is indistinguishable from that ceiling (0.71), and roughly half of the run-to-run noise comes from the initial structure. In simulations with the measured noise, giving further runs only to candidates whose membership of the top set is uncertain matches four uniform replicates with 1.8 runs per candidate for the top 10% and 1.2 for the top 1%; a check on real runs is consistent but underpowered. The lithium

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and anion diffusivities and the ionic conductivity behave alike. A decision guide and all code are provided.

Keywords: molecular dynamics, simulation output analysis, uncertainty quantification, label noise, simulation budget allocation, high-throughput screening, polymer electrolytes

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1. Introduction

Molecular dynamics (MD) is increasingly used not to study one material carefully but to label thousands of candidates, either to rank them directly or to train machine-learning models that extend the search [1]. Solid polymer electrolytes are a representative case: lithium transport is slaved to the slow segmental dynamics of the polymer [2, 3], so the simulations needed to converge transport coefficients are long, while the chemical space is large. The HTP-MD platform, the largest public collection of such simulations, distributes thousands of short runs of 5 ns and a smaller number of 50 ns runs, with raw trajectories and derived properties [1, 4]. Each label in such a collection is the output of a stochastic simulation, and how to spend computation on it is a question of computational science as much as of chemistry.

A label obtained this way has two defects that are easy to overlook once it sits in a table. It is *time dependent*: the self-diffusion coefficient is defined in the long-time limit, a diffusivity extracted before the mean-squared displacement (MSD) becomes linear is a property of the observation window [5], and the window of a screening run is short. It is also *noisy*: the amorphous structure is sampled once, and two runs of the same polymer from different initial structures can disagree substantially [1]. The dataset's authors addressed both defects by training a multi-task graph network on many short and few long runs, assuming noise independent of the polymer [1]. Learning from noisy labels is well studied in machine learning [6], and the statistics of diffusion estimates from finite trajectories are well understood [7–10], as are best practices for quantifying the sampling uncertainty of MD observables [11]. That measurement noise bounds the performance any model can show is also established: for experimental activity data, the replicate variability sets the maximum correlation or R^2 a model can reach [12, 13], and computational neuroscience estimates the same bound as a noise ceiling [14]. In the language of stochastic simulation, the remaining questions are classical problems of output analysis: the bias left by an incomplete warm-up [15, 16],

the choice between independent replications and longer runs [17], and the selection of the best of many noisy systems under a budget [18, 19]. For simulated labels, however, to our knowledge the noise across a real screening campaign has not been measured, although the same question has been raised for chemistry datasets in general [20]; its sources have not been separated, whether a model actually reaches the resulting ceiling has not been tested, and the noise has not been used to decide how a fixed simulation budget should be spent. Unlike experimental noise, simulation noise can be reduced by spending more computation, which turns the ceiling into a question of allocation.

We answer these questions on HTP-MD without training a corrective model, using three features of the public collection: 94 screening systems simulated twice from different initial structures, 133 systems simulated at both lengths, and the raw long trajectories, which can be truncated to emulate shorter runs. Our contributions are:

1. **Time dependence of the labels.** A census of all 6,270 trajectories, the lag dependence of the apparent diffusivity in the long runs, and a slow decrease of lithium mobility within each run that makes its beginning faster than its later part and follows an incomplete relaxation of the polymer.
2. **Reliability and noise ceiling.** The run-to-run noise of a short label, its structural and dynamic sources and its dependence on mobility, the ceiling it imposes on the performance that any predictor can show against such labels, a machine-learning model that comes close to that ceiling, and the ability of one short run to select the best candidates, checked on two further properties, the anion diffusivity and the ionic conductivity.
3. **Spending the budget.** An evaluation of estimators and stopping rules on truncated trajectories, and an evaluation, with the measured noise, of known adaptive strategies that concentrate replicates on the candidates near the selection cut-off.
4. **A decision guide and reproducible code** for each screening purpose.

The tools come from measurement theory and simulation optimisation. Applied together to a real screening campaign, they reveal an initialization bias shared by all short labels, a structural share of the noise that no extension of a run can remove, and a ceiling that no model can exceed against single-run labels, and they turn these quantities into a budget rule that a new campaign can apply after replicating a small random subset of its candidates.

2. Methods

2.1. Data set

We use every trajectory listed by the public HTP-MD web application [4] on 6 to 9 October 2026: 6,270 trajectories of amorphous polymer/LiTFSI systems simulated with the PCFF+ force field at 353 K, 6,137 for 5 ns and 133 for 50 ns (Table S1). For each trajectory the platform distributes the MSD curves of Li^+ and of the anion at 100 lags, derived transport properties and the raw trajectory, with frames every 2 ps. The storage paths identify each system. Every long run has a 5 ns run of the same system (96 screening polymers and 37 taken from the literature), and, as in the original study [1], the two runs of such a pair start from the same initial amorphous structure. In addition, 94 screening systems were simulated twice for 5 ns from different initial structures. The storage paths label these second runs as a random sample of 100 screening systems (`rand_100`), and 92 of the 96 long screening runs belong to the same systems. Neither subset is selected by mobility: the short-run labels of the replicated systems and of the systems with a long run are distributed as those of the rest of the screening set (Kolmogorov-Smirnov $p = 0.61$ and 0.65; Levene $p = 0.82$ and 0.65), so the noise they measure applies to the whole screening population. Only the 37 literature polymers were chosen deliberately, and they are analysed separately. Every system contains 50 LiTFSI ion pairs at a nominal molality of $1.5\,\mathrm{mol\,kg^{-1}}$, equilibrated for 5 ns before production [1]; from the starting structures, the systems have 2,328 to 7,541 atoms (median 6,002) in 1 to 15 polymer chains (median 9), in cubic boxes of 3.0 to 4.2 nm side (ion and chain counts verified on 125 complete structure files).

2.2. Labels and their time dependence

For an MSD curve averaged over time origins and ions, $\mathrm{MSD}(\tau) = \langle|\mathbf{r}_i(t+\tau) - \mathbf{r}_i(t)|^2\rangle_{i,t}$, we define the apparent diffusivity $D_{\mathrm{app}}(\tau) = \mathrm{MSD}(\tau)/6\tau$. The diffusivity distributed by the platform is the *end-point* estimate $D_{\mathrm{end}}(T) = \langle|\mathbf{r}_i(T) - \mathbf{r}_i(0)|^2\rangle_i/6T$, which uses only the first and last frames [21]; we also use the origin-averaged $D_{\mathrm{app}}(T/2)$. Diffusion corresponds to a lag-independent D_{app} and an MSD exponent $\beta = \mathrm{d}\ln\mathrm{MSD}/\mathrm{d}\ln\tau = 1$. We estimate an effective exponent by a least-squares fit of $\ln\mathrm{MSD}$ against $\ln\tau$ over lags between 20% and 60% of the run, and call a run diffusive when $\beta \geq 0.9$; this is an operational classification, and its sensitivity to the window is reported in Table S2. For the long runs we compute $D_{\mathrm{app}}(\tau)$ from

0.5 ns to 25 ns and report its ratio to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ and the Spearman correlation of the rankings. The reference is a finite-lag quantity, not a converged coefficient. To test whether the beginning of a run differs from its later part, we split each long run into ten consecutive 5 ns segments and compute, from the data of each segment alone, the end-point and origin-averaged estimates and the effective exponent; each segment is compared with the geometric mean of the segments between 25 and 50 ns. In the same segments we follow the mobility of the anion and of polymer atoms, the potential energy and pressure (production is at constant volume), and the lithium-anion clusters computed during the simulations (Supplementary Material).

2.3. Noise model, reliability and noise ceiling

All noise statistics are computed on $\ln D$. Following Ref. [1], we model each label as a system-level value plus run-specific noise. Here, the system-level value is the expected 5 ns label across initial structures and dynamical realizations. It can include biases shared by short runs, such as the early slowdown reported below; it is not the long-time transport coefficient. Thus, the reliability and noise ceiling describe how well short-run labels can be predicted. A separate, system-dependent bias distinguishes those labels from the long-time property. For the replicated systems, the standard deviation of the difference between the two runs divided by $\sqrt{2}$ estimates the noise σ_{run} of one run. With σ_{tot} the spread of all short runs, the reliability of one run is

$$r = 1 - \sigma^2_{\mathrm{run}}/\sigma^2_{\mathrm{tot}}, \tag{1}$$

the fraction of the variance between candidates that reflects differences between systems. We also report the one-way intraclass correlation computed within the replicated subset. A predictor that recovered the property of every system exactly would correlate with one label at most $\sqrt{r}$ and explain at most a fraction r of its variance, the bound used for experimental data [12, 13] and known as the noise ceiling in computational neuroscience [14]; the mean of n runs has reliability $nr/[1 + (n - 1)r]$ [22, 23]. To translate reliability into selection we simulate Gaussian populations with the measured variances and record the fraction of the truly best 5%, 10% and 20% recovered when selecting by the mean of n runs.

Two sources of noise are separated under an additive model with independent structural and dynamic contributions, so the split is a statistical attribution conditional on that model rather than a causal decomposition.

Replicas start from different structures, so their difference contains structural and dynamic noise. The 5 ns run of a pair and the first 5 ns of its long run start from the same structure, so their difference contains only dynamic noise, σ_{dyn}; the two are distinct dynamical realisations (their labels are not identical), although the platform does not document whether they share any part of the equilibration. Then $\sigma^2_{\mathrm{str}} = \sigma^2_{\mathrm{run}} - \sigma^2_{\mathrm{dyn}}$, with σ_{run} from the 94 replicated systems and σ_{dyn} from the 96 screening pairs. The short run of a pair is also the original replica of that system, so the two differences share a run; we therefore repeat the split on the 92 systems that have both, resampling them jointly. We test whether the noise depends on mobility by the rank correlation between the absolute difference of two replicas and their mean. Intervals throughout are obtained by bootstrap over systems (2000 resamples).

2.4. Machine-learning demonstration

To compare a real model with the ceiling we predict the end-point $\ln D$ of the original 5 ns run of each system from its monomer alone. The input is the Morgan count fingerprint (radius 2, 2048 bits; RDKit) of the monomer SMILES distributed by HTP-MD, including the dummy atoms that mark the connection points, so chain architecture is not represented. The model is a random forest (scikit-learn; 400 trees, a fraction 0.3 of the features per split, at least two samples per leaf), with these settings fixed in advance and not tuned. It is trained on the 6,042 systems that have an original screening run, literature polymers included (1 system appears only as a replica and is left out), and evaluated by 5-fold cross-validation split by system, so that every prediction is out of sample. As a stricter split we group the monomers into chemical families by Butina clustering of the Tanimoto distances of their binary Morgan fingerprints, at two similarity thresholds, and keep each family within one fold. Intervals for the correlations are obtained by bootstrap over systems of the out-of-fold predictions. To measure the effect of splitting by trajectory instead of by system, we repeat the cross-validation with every trajectory, including the second runs of the replicated systems, as its own group, and compare the two splits on the replicated pairs whose two runs fell in different folds (84 of the 94; in the other 10 both runs were in the same test fold).

2.5. Selection from paired runs

For the 133 pairs we compare the end-point label of the 5 ns run with the reference of the long run of the same system: median ratio, Spearman correlation (separately for screening and literature polymers) and the recall of the best 5%, 10% and 20% systems, that is, the fraction of the k best according to the reference that are also among the k best according to the short run, against the recall k/n of a random choice. Because the two runs share their initial structure, this comparison contains only the dynamic part of the noise.

2.6. Truncated trajectories and stopping rules

From the raw trajectory of each long run we extract the unwrapped lithium positions, subtract the centre-of-mass motion of the whole system as the released library does [21] (validation in the Supplementary Material), truncate the trajectory to its first T nanoseconds and estimate the diffusivity from that prefix alone with four estimators: the end-point, the origin-averaged $D_{\mathrm{app}}(T/2)$, one sixth of the slope of a linear fit to the MSD over $0.2T$ to $0.5T$, the conventional procedure [5], and a power law MSD $= A\tau^{\beta}$ fitted over $0.1T$ to $0.5T$ and extrapolated to τ_{ref}, $\widehat{D_{\mathrm{app}}} = A\tau_{\mathrm{ref}}^{\beta-1}/6$. Estimates are compared with the reference of the full run through the typical multiplicative error $\exp(\mathrm{median}\,|\ln(\hat{D}/D_{\mathrm{ref}})|)$; because the reference shares data with the prefix, we repeat the comparison against references at other lags and against the late slope of the MSD. Stopping rules stop a run when its effective exponent reaches a threshold or when the bootstrap coefficient of variation of an estimate falls below a tolerance; runs that never stop are simulated to 50 ns, and cost and error are averaged over all runs. Thresholds are chosen on a random half of the systems and evaluated on the other half (500 partitions).

2.7. Adaptive allocation of replicates

Consider a campaign of $N = 6{,}137$ candidates, a hypothetical population the size of the set of short runs, with a budget of b 5 ns runs per candidate on average, whose goal is the top k, a ranking-and-selection problem [18]. System values are drawn from $\mathcal{N}(0, \sigma^2_{\mathrm{sys}})$ with $\sigma^2_{\mathrm{sys}} = r\sigma^2_{\mathrm{run}}/(1-r)$, and each run adds noise whose standard deviation varies linearly with mobility so as to reproduce the measured noise of the slow and fast halves; every replicate starts from a new structure and carries the full noise. The unit

of cost is therefore a complete new-structure run, which in the HTP-MD protocol comprises 5 ns of equilibration and 5 ns of production; budgets and savings below are counted in such runs. A normal prior and a Gaussian approximation of the posterior, with the noise evaluated at the observed mean of each candidate rather than at its unknown value, give the mean and uncertainty of each candidate. We compare three allocations of the budget. *Uniform*: b runs for every candidate. *Successive halving* [24]: one run for all, then further runs for the best $4k$, $2k$ and k, in rounds that restart from $4k$ until the budget is spent. *Boundary-focused*: one run for all, then batches of 5% of N further runs to the candidates with the largest $p(1-p)$, where p is the posterior probability of lying above the current top-k cut-off (the k-th largest posterior mean, treated as fixed within a batch), in the spirit of optimal computing budget allocation [19] and of its version for selecting a subset of the best designs [25]. The final selection is the top k by posterior mean, and the metric is the recall of the true top k, averaged over 200 simulated campaigns (Monte Carlo standard error at most 0.4 percentage points). As a simple reference we add a *two-stage re-screening*: one run for all, then the rest of the budget spread evenly over a fixed shortlist of the best $2k$, $4k$ or $8k$ after the first run, taking for each budget the best of the three shortlists. Robustness is checked with the empirical shape of the short-run distribution instead of a normal one, with the origin-averaged label, with a constant instead of a mobility-dependent noise, with a misestimated reliability (the runs follow the measured noise, but the allocation and the selection use the ends of its 95% interval), and with the true reliability at either end of that interval. The algorithm and its parameters are given in the Supplementary Material.

The public data allow a partial check on real runs. In the replicated systems, the first run is the original screening run and the second starts from a new structure, and 92 of them also have a long run, which starts from the same structure as the first. We use them in two ways, with noise parameters estimated leaving out the system being evaluated. First, from one run of each system the model predicts the probability that the other run exceeds the top-k cut-off; we compare these predictions with what happens (Brier score against the base rate and calibration slope). Second, we run a small real campaign: every system receives its first run, and 46 second runs go either to the candidates with the largest $p(1-p)$ or to randomly chosen ones; the selection is evaluated against the long runs, which are independent of the second runs.

3. Results and Discussion

3.1. Labels depend on the simulated time

After 5 ns the median effective exponent of the lithium MSD is 0.43 (10th to 90th percentile range 0.26 to 0.60): 97.8% of the runs have $\beta < 0.7$ and 2 of 6,137 meet the diffusive criterion (Fig. 1). Ten times more simulation moves the distribution but does not resolve the problem: at 50 ns the median is 0.65 and 5 of 133 runs are diffusive (0 to 19 depending on the fitting window, Table S2). In the long runs the apparent diffusivity falls steadily with the lag (Fig. 2): at 5 ns it is a median 1.95 times its value at 25 ns (10th to 90th percentile range 1.47 to 2.40), and only 73.7% of the systems are within 25% of the reference at 15 ns (Table S4). Rankings are far more stable: the Spearman correlation with the ranking at 25 ns is 0.91 already at 0.5 ns and 0.96 at 5 ns. A short-run label is therefore an apparent, window-dependent quantity whose value should not be compared across studies with different lengths, but whose ordering carries information.

Labels also depend on *when* the time is simulated (Fig. 3). Analysed with its own data only, the first 5 ns segment of a long run is a median 1.63 times faster than the segments between 25 and 50 ns (95% interval 1.52 to 1.81; 1.47 with the origin-averaged estimator), and it is faster in 93% of the systems (Wilcoxon signed-rank $p = 3.6 \times 10^{-22}$). The slowdown is not a short transient: the apparent diffusivity keeps falling by a factor of 0.92 every 10 ns between 5 and 50 ns, in 85% of the systems, and the effective exponent of the segments decreases from 0.48 in the first to 0.31 in the late ones. Lithium mobility therefore keeps decreasing throughout the runs, despite the 5 ns equilibration that precedes production [1], and the whole system is still relaxing (Table S5). At constant volume, the potential energy falls between the first and the late segments in all 133 systems, by a median 0.014 kcal mol^{-1} per atom, 6.6 times the fluctuation of its 100 ps averages, and the anion and the polymer slow down with lithium ($\times$1.46 and $\times$1.43). Across systems, the slowdown of lithium follows that of the polymer (Spearman 0.68, 95% interval 0.57 to 0.76) and the relaxation of the energy (0.58), but not the change in ion clustering, which is small and uncorrelated with it (-0.03). The early, faster window is thus consistent with a polymer that has not finished relaxing after the equilibration, as expected if lithium transport is slaved to segmental dynamics [2, 3]; these are correlations, not a demonstration of cause. In simulation terms this is an initialization bias that the fixed equilibration does not remove. Detecting the end of equilibration from the

trajectory itself [26], as truncation heuristics do for the warm-up of other stochastic simulations [15, 16], would be a natural way to start production per system. The effect is stronger for the screening polymers (1.80) than for those taken from the literature (1.28).

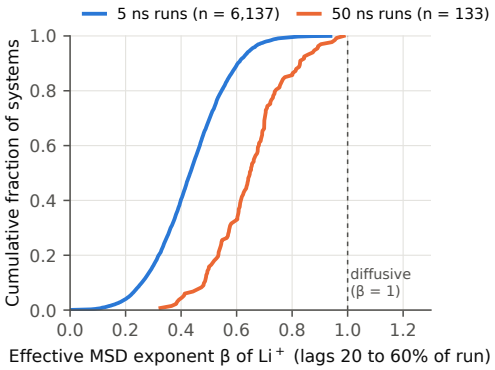


Figure 1: Cumulative distribution of the effective MSD exponent β of Li^+ in the 5 ns and 50 ns runs. The dashed line marks diffusive motion ($\beta = 1$); runs are counted as diffusive when $\beta \geq 0.9$.

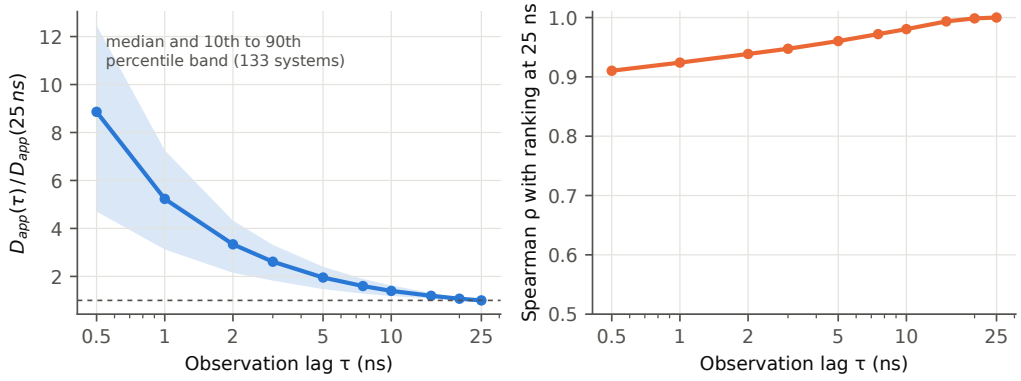


Figure 2: Lag dependence in the 133 runs of 50 ns. Left: apparent diffusivity of Li^+ relative to its value at 25 ns (median and 10th to 90th percentile band). Right: Spearman correlation between the rankings at lag τ and at 25 ns.

3.2. Run-to-run noise and its sources

A single 5 ns run deviates from the value of its system by a typical factor of 1.43 for the end-point label ($\sigma_{\mathrm{run}} = 0.35$ in log units) and 1.32 for the

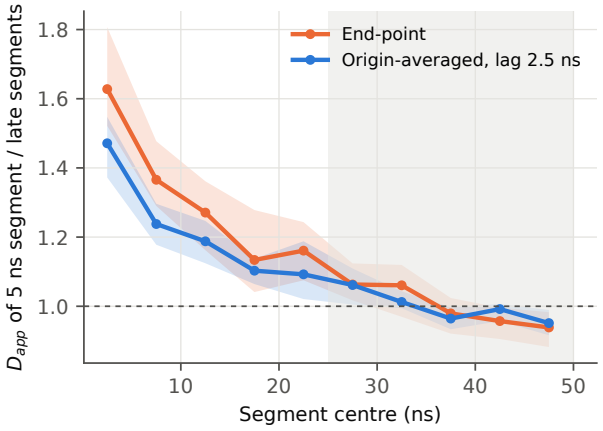


Figure 3: Apparent diffusivity of consecutive 5 ns segments of the 133 long runs, each computed from its own data, relative to the segments between 25 and 50 ns (shaded): median over systems and 95% bootstrap interval.

origin-averaged one, estimated from pairs of runs of the same system from different structures (Table S6). Against the spread of the whole screening set, the reliability of one end-point label is 0.47 (95% interval 0.31 to 0.62), and 0.58 (0.44 to 0.70) for the origin-averaged label; the intraclass correlation within the replicated subset agrees (0.53 and 0.62). Roughly half of the variance between screened candidates is therefore noise.

The noise has two comparable sources. The 5 ns run of each pair agrees with the first 5 ns of its long run, which starts from the same structure, to a median ratio of 1.00 with a noise of 0.24, against 0.35 between replicas from different structures: the initial structure accounts for 55% (31 to 72%) of the run-to-run variance and the dynamics for the rest; on the 92 systems with both comparisons, resampled jointly, the share is 57% (44 to 69%). The originals, the replicas and the long runs were produced in different batches of the data set, but none is shifted or differently spread (Supplementary Material, after Table S6). Re-sampling the structure and averaging the dynamics both reduce the noise. A replicate that starts from a new structure samples both sources, as ensembles of independent runs do in other MD applications [27], but it also costs a new equilibration; this is the classical choice between independent replications and one longer run [17]. The measured split shows why that cost can be worth paying: even if extending a single run removed the dynamic noise completely, the structural part would remain, and the

reliability of a label from one structure could not exceed 0.62, whereas the mean of two runs from different structures already reaches 0.64. The bound assumes that the memory of the initial structure does not fade as the run is extended, which these data cannot test, and a longer run also measures a later, slower window. In the allocation results below every run, uniform or adaptive, starts from a new structure. The noise also depends on mobility: it is 0.40 in the slower half of the replicated systems and 0.30 in the faster half (rank correlation between the absolute difference and the level -0.26, $p = 0.01$), whereas the learning model of the original study assumes noise independent of the polymer [1]. Two candidates measured once each differ by a factor of 1.65 from noise alone at one standard deviation and up to 2.67 at the 95% level.

The structural share is consistent with how slowly the local environment of lithium renews. In the 133 long runs, a lithium ion has a median of 2.0 of the polymer atoms followed here within 3.4 Å in the screening polymers and 3.4 in the literature polymers, and 54% and 66% of those contacts are still present after 5 ns (Table S7). These atoms are of the types the platform lists as solvating but are not always the ones that coordinate lithium, so the contacts describe the local polymer environment rather than the coordination shell. A short run therefore samples essentially the environment it starts from, and a new initial structure brings a new one. Ion aggregation, read from the cluster populations of the platform, matters more for the bias of short labels: systems with more lithium in large clusters are slower (rank correlation -0.25, 95% interval -0.41 to -0.08), are more overestimated by the short run (0.26, 0.10 to 0.42) and are ranked worse by it (0.18, 0.02 to 0.34), consistent with cluster rearrangement being one of the slow processes that 5 ns does not reach. No descriptor of the environment or of aggregation predicts the replica noise of a system (all rank correlations within ± 0.18), so a single reliability for the whole screening population, as used below, is not hiding a chemistry with markedly noisier labels.

3.3. A noise ceiling for learned predictors

The reliability bounds what any predictor can achieve against short-run labels (Table 1). A model that recovered the property of every system exactly would correlate with a single end-point label at most 0.69 (95% interval 0.56 to 0.79) and explain at most $R^2 = 0.47$ of its variance; with the origin-averaged label the ceilings are 0.76 and 0.58. Choosing the best 10% by one end-point label recovers 45% of the truly best tenth, which is also the

apparent recall of a perfect model evaluated against such labels. Averaging runs raises the ceiling: two end-point runs give a reliability of 0.64, four give 0.78, and a reliability of 0.8 requires 5 runs (3 with the origin-averaged label). The original study estimated that its graph network predicts the conductivity with the accuracy of about four simulations per polymer [1]; for the lithium diffusivity studied here, four runs correspond to a reliability of 0.78.

A real model comes close to this ceiling (Table 2). A random forest on Morgan fingerprints of the monomer, trained on the 6,042 systems with cross-validation split by system (see Methods), gives out-of-sample predictions that correlate with the labels with Pearson $r = 0.71$ (95% interval 0.70 to 0.72; $R^2 = 0.50$), statistically indistinguishable from the estimated ceiling of 0.69 (95% interval 0.56 to 0.79): the ratio of the two is 1.03, and 0.90 to 1.27 across the interval of the ceiling, whose uncertainty dominates. Keeping each chemical family of monomers within one fold barely changes the result: 0.70 with 1,415 families (Tanimoto ≥ 0.6) and 0.70 with 144 (Tanimoto ≥ 0.4), although the screened chemistry is narrow: even then the most similar training monomer has a median similarity of 0.65, against 0.72 when splitting by system, so this tests interpolation within the screened chemistry, not generalisation to new families. That the point estimate exceeds the ceiling slightly suggests that the reliability of the whole population is somewhat higher than the replicated subset indicates; either way, a simple model is already close to what single short-run labels allow. Being this close to the estimated ceiling makes further improvements by better architectures hard to demonstrate against single short-run labels, although it does not exclude them; averaged labels or long-run references would show them more clearly. The replicated systems are consistent with the measurement model. For the same predictions, the correlation rises from 0.67 and 0.69 against either single label to 0.78 against the mean of the two, as the reliability of a two-run mean predicts (0.79; gain 95% interval 0.07 to 0.13). Against such labels, better labels rather than better models are what can raise the measured performance.

Against the long runs, the model, which never saw these systems, ranks the 50 ns reference as well as the system's own 5 ns run among the screening polymers (Spearman 0.67 against 0.68), but worse among the literature polymers (0.67 against 0.93), whose chemistry lies outside the screened space. Because each long run shares its initial structure with the short run, this comparison favours the short run; a model trained on short labels is thus a substitute for a short simulation within the chemical space it was trained on,

not beyond it.

Two practical consequences follow. A test metric computed against single short-run labels cannot exceed the ceiling, so a model close to it is as good as the labels allow, and progress has to be measured against averaged labels. And because 94 systems appear twice in the public dataset, splits should be made by system; in this dataset the effect of splitting by trajectory instead is small (0.66 against 0.68 on the 168 trajectories of the 84 replicated pairs split across folds), because the model agrees with either run of a replicated system more closely than the two runs agree with each other (rank correlation 0.48), so the partner's label adds little.

Table 1: Noise ceiling of short-run labels in the screening population: reliability of the mean of n runs, maximum correlation of a perfect predictor with that label, and recall of the truly best systems when selecting by it.

				Recall of the true top		
Label	Runs averaged	Reliability	Max. correlation	5%	10%	20%
End-point	1	0.47	0.69	38%	45%	56%
	2	0.64	0.80	49%	56%	64%
	3	0.73	0.85	56%	63%	70%
	4	0.78	0.88	61%	66%	73%
Origin-averaged, lag $T/2$	1	0.58	0.76	45%	52%	61%
	2	0.73	0.85	56%	63%	70%
	3	0.80	0.90	63%	68%	75%
	4	0.84	0.92	67%	72%	77%

Table 2: Out-of-sample correlation of a random forest on monomer fingerprints with short-run labels (cross-validation split by system, or by chemical family of the monomer), against the noise ceiling of each label. The ceilings and their intervals rest on the run-to-run noise of the 94 replicated systems, not on the 6,042 systems of the first rows.

Target of the out-of-sample prediction	Systems	Pearson r (95% CI)	Ceiling (95% CI)
One 5 ns label, all systems	6,042	0.71 (0.70 to 0.72)	0.69 (0.56 to 0.79)
split by fine chemical family (Tanimoto ≥ 0.6)	6,042	0.70 (0.69 to 0.71)	0.69 (0.56 to 0.79)
split by coarse chemical family (Tanimoto ≥ 0.4)	6,042	0.70 (0.69 to 0.72)	0.69 (0.56 to 0.79)
One 5 ns label (first / second run), replicated systems	94	0.67 / 0.69	0.69 (0.56 to 0.79)
Mean of two 5 ns labels, replicated systems	94	0.78 (0.66 to 0.86)	0.80 (0.69 to 0.88)

3.4. Selecting candidates from one short run

For the 133 systems simulated at both lengths, the end-point label of the 5 ns run is a median 2.90 times the long-run reference (10th to 90th percentile range 1.58 to 4.74; Fig. S2), and the Spearman correlation between them is 0.84 (95% interval 0.76 to 0.89). The correlation owes much to the wide range of mobilities in this subset: it is 0.93 among the 37 literature polymers but 0.68 among the 96 screening polymers. In the whole paired set the short run identifies 11 of the 13 best systems (Table 3), but these top sets consist almost entirely of literature polymers (13 of 13 and 24 of 27), which are easy to separate. Within the 96 screening polymers the short run still recovers 7 of the 10 best and 14 of the 19 best, against 10% and 20% by chance. These recalls are optimistic, because the short and long runs of a pair share their initial structure and only the dynamic part of the noise separates them. A short run thus separates good from poor electrolytes but cannot resolve similar ones, in line with the noise ceiling above.

The median ratio of 2.90 is larger than the 1.95 obtained at the same lag from the origin-averaged long curves, and this slowdown explains the gap. The short runs behave like the first 5 ns of the long runs (median ratio 1.00), and that first segment is 1.42 times faster than the average of all segments, which is roughly what separates a single early window from a curve averaged over all time origins. Short labels in the dataset are therefore systematically biased upwards by the early, faster part of the trajectory, and more so for the screening polymers. For ranking, this bias matters only to the extent that it differs between systems.

3.5. Spending the budget within a run: estimators and stopping

When only the first T nanoseconds of a long run are available (Fig. 4, Table S8), the end-point, half-lag and linear-fit estimates overestimate the reference at every short length, by typical factors of 2.83, 4.05 and 2.02 at 5 ns, and their bootstrap intervals seldom contain it (3%, 0% and 39% of the runs): a within-run error bar describes the apparent diffusivity at the lags available in the prefix, not its distance to the long-lag value. A power-law extrapolation removes most of the systematic deviation (median ratio 0.95, typical error 1.37 from 5 ns) but ranks the systems slightly worse (Spearman 0.75 against 0.81 for the end-point), because it extrapolates an exponent that is not reproducible between runs, and it underestimates the late slope of the MSD (median ratio 0.54; Table S9). A power law with $\beta < 1$ has no crossover to diffusion: its slope keeps falling with the lag, whereas the

Table 3: Recall of the best systems according to the 50 ns reference when they are selected from the 5 ns run of the same system (133 systems; the last rows only the 96 screening polymers). Intervals by bootstrap over systems; "Random" is the expected recall of a random selection of k systems. All recalls are optimistic, because the two runs of a pair share their initial structure.

Estimator, 5 ns run	Top	k	Hits	Recall (95% CI)	Random
End-point	5%	7	7	100% (57 to 100)	5%
	10%	13	11	85% (62 to 100)	10%
	20%	27	23	85% (70 to 93)	20%
Origin-averaged, lag $T/2$	5%	7	6	86% (57 to 100)	5%
	10%	13	10	77% (62 to 100)	10%
	20%	27	22	81% (70 to 93)	20%
End-point, screening only	5%	5	4	80% (40 to 100)	5%
	10%	10	7	70% (50 to 90)	10%
	20%	19	14	74% (47 to 89)	20%

true MSD must eventually become linear. The extrapolation is therefore an empirical estimate of the apparent diffusivity at a chosen lag within or near the observed window, not of the long-time coefficient. Adaptive stopping rules, based on the exponent or on the bootstrap precision, do not improve on the cost-error frontier of fixed-length runs: they either rarely stop early or stop early at the price of large errors (Fig. S3 and Table S10), and settings chosen on half of the systems often miss their target on the other half: the power-law length chosen for a typical error of $\times 1.3$ meets it on the held-out systems in only 45% of the partitions (Table S11). Within a run, then, there is little to save; the budget is better spent across candidates.

3.6. Spending the budget across candidates: adaptive allocation

The noise model measured above is compatible with the replicated systems: the differences between replicas show no departure from a Gaussian (excess kurtosis 0.13, Shapiro-Wilk $p = 0.81$). With it, how replicates are distributed matters more than how many are run (Fig. 5; values in Table S12). Uniform replicates improve slowly: for the top 10% of 6,137 candidates the recall rises from 48% with one run to 72% with four. Boundary-focused allocation reaches 75% with two runs per candidate on average and 84% with four, and matches four uniform replicates with 1.8 runs per candidate, a saving of $\times 2.2$. The saving grows with the selectivity of the target: $\times 2.7$ for

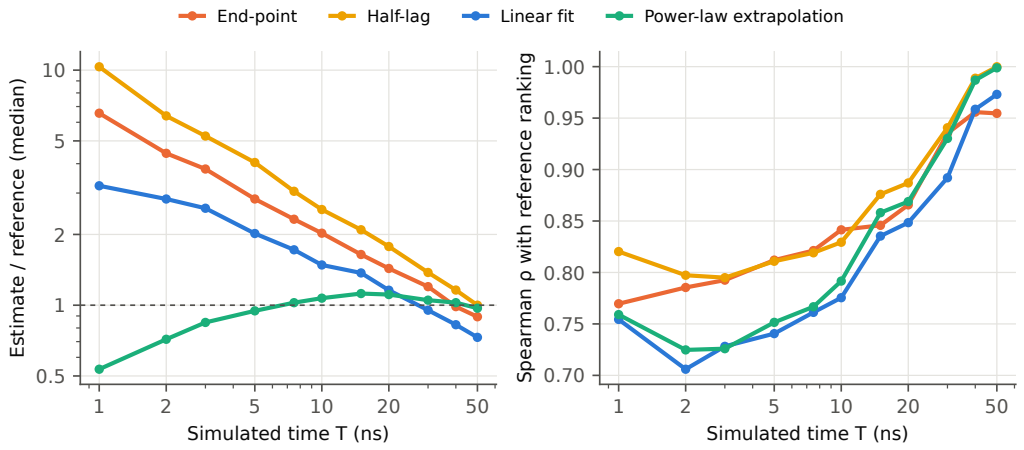


Figure 4: Estimators computed from the first T nanoseconds of the 133 long runs. Left: median ratio to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ of the full run. Right: Spearman correlation with the reference ranking.

the top 5% and ×3.2 for the top 1% (Table S13). Successive halving, which re-runs the current leaders in repeated rounds, performs similarly: with the end-point label it is better for the top 1% at small budgets (74% against 68% with 1.25 runs per candidate), and with the origin-averaged label it is worse at larger budgets (84% against 89% with four). Even a two-stage re-screening, in which the budget left after the first run is spread evenly over a fixed shortlist of the best candidates, captures most of the gain: with a shortlist of the best $4k$ fixed in advance it saves ×3.0 for the top 1%, ×2.4 for the top 5% and ×1.8 for the top 10%, and the best of three shortlist sizes, chosen in hindsight, adds little (×3.2, ×2.4 and ×1.8). Most of the saving therefore comes from not spending replicates on candidates that the first run already places far from the cut-off; allocating them where the selection is still in doubt adds the rest, and does so without a shortlist size to tune. The price is sequential work: in batches of 5% of the candidates, matching four uniform replicates takes 17 rounds of simulation and analysis for the top 10% and 5 for the top 1%, against a single extra round for the two-stage re-screening. Where each round waits in a queue, the two-stage scheme is the practical choice, and its loss is small for the most selective targets. Nor does the saving depend on knowing the reliability exactly: planning with $r = 0.31$ or 0.62 instead of 0.47 gives savings of ×2.2 and ×2.1 for the top 10% (Table S14), and if the true reliability lies at either end of its interval

the saving is ×1.8 to ×2.5 for the top 10% and ×2.4 to ×3.2 for the top 1%. The result holds with the empirical distribution of the short runs and with the origin-averaged label, and knowing that slow systems are noisier does not improve the allocation (Supplementary Material). In practice, a campaign should give one short run to every candidate and spend the rest of the budget on new-structure replicates of the candidates whose membership of the top set is uncertain.

These gains are predictions of the noise model, since the public data contain at most two runs per system; the replicated systems allow only a partial check, which is consistent with them but has little statistical power (Table S15). The model's probabilities of exceeding the top-10% cut-off are informative on real runs, improving the Brier score of the base rate by 36% with a calibration slope of 1.10 (95% interval 0.66 to 2.27). In the real campaign on 92 systems, sending the 46 second runs to the boundary recovers 78% of the top 10% according to the long runs, as many as giving a second run to every system (78%), against 71% for randomly placed second runs and 67% with first runs only. With 9 systems in the top set the difference from random placement is not significant (95% interval from −11 to +22 percentage points), and for the top 20% no strategy improves on the first run alone. The long runs share their structure with the first runs, which probably favours the first run over any second run from a new structure, by an amount we cannot quantify. The calibration check, in turn, concerns a second noisy run rather than the latent value that the allocation targets. The real runs therefore do not contradict the simulated gains, but they cannot show that adaptive allocation beats random placement.

3.7. Other properties: anion diffusivity and ionic conductivity

To test whether these conclusions depend on the property, we repeated the main analyses for the diffusivity of the TFSI anion and for the ionic conductivity distributed by the platform (Table 4), which is the property the original study set out to predict [1]. That conductivity is a cluster Nernst-Einstein estimate [28]: it combines the end-point diffusivities of both ions with the populations of ionic clusters, so it includes ion pairing (we reproduced the distributed values from the trajectories to within a constant unit factor). Because the platform distributes only the origin-averaged estimate for the anion, both ions are compared with that estimator. The anion label is much less noisy: its reliability is 0.80 (95% interval 0.75 to 0.85) against 0.58 for lithium, so its noise ceiling is higher (0.89 against 0.76) and a short

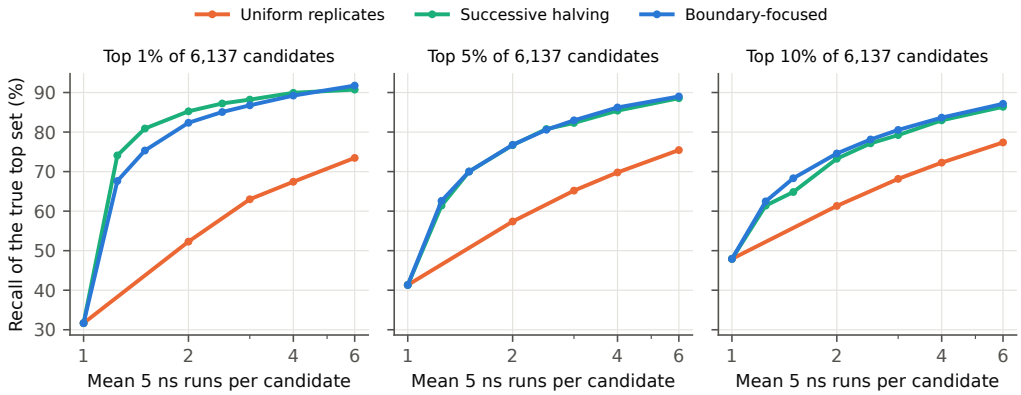


Figure 5: Recall of the true top set as a function of the mean number of 5 ns runs per candidate, for three ways of allocating the budget (200 simulated campaigns of 6,137 candidates with the measured noise; each run starts from a new, equilibrated structure). The true top set is the top k by the expected 5 ns label of each system, not by its long-time diffusivity.

run ranks the screening polymers better against the long runs (Spearman 0.83 against 0.70). The structure of the results, however, is the same. The random forest comes close to the ceiling for both ions (0.86 and 0.77), its correlation with the mean of two replicas rises as the measurement model predicts (0.90 against an expected 0.90 for the anion), and adaptive allocation matches four uniform replicates with less than half the runs ($\times$2.7 and $\times$2.3 for the top 10%). The conductivity behaves in the same way: its reliability is 0.65 (0.53 to 0.76), its ceiling 0.81, and adaptive allocation saves $\times$2.4. One difference is chemical. The random forest falls short of the conductivity ceiling (0.73 against 0.81), whereas it comes close to those of both diffusivities: monomer fingerprints describe the mobility of the ions better than how they pair. And ranking by lithium mobility is not ranking by conductivity. Across the screened systems the two labels of the same run have a rank correlation of only 0.56 (0.66 for the long runs), and the best 10% by lithium diffusivity contain 47% of the best 10% by conductivity, although the noise of the two labels is largely shared (correlation 0.78 between the replica differences). The choice of label is therefore a decision about chemistry, not only about noise: a campaign that looks for good conductors should screen the conductivity, with the same noise analysis. The numbers are specific to each property, but for these three properties the conclusions are the same.

Table 4: Main results for the lithium and the TFSI anion diffusivities, both with the origin-averaged estimator at half the run length, and for the cluster Nernst-Einstein conductivity distributed by the platform. Noise statistics from the 94 replicated systems; random forest and allocation as in the previous sections (allocation with constant noise).

	Li^+	$TFSI^-$	Conductivity
Run-to-run noise (log)	0.28	0.19	0.31
Noise, slow / fast half	0.29 / 0.26	0.20 / 0.18	0.31 / 0.31
Reliability (95% CI)	0.58 (0.44 to 0.70)	0.80 (0.75 to 0.85)	0.65 (0.53 to 0.76)
Noise ceiling	0.76	0.89	0.81
Random forest vs one label	0.77	0.86	0.73
Random forest vs two-run mean (expected)	0.84 (0.86)	0.90 (0.90)	0.78 (0.76)
Spearman with 50 ns run, screening	0.70	0.83	0.77
Spearman with 50 ns run, literature	0.94	0.96	0.98
Runs matching 4 uniform replicates (top 10%)	1.7 (×2.3)	1.5 (×2.7)	1.7 (×2.4)

3.8. Practical guidance

Table 5 turns these results into recommendations, ordered from the least to the most demanding purpose. They apply to the setting studied (LiTFSI in amorphous polymers, PCFF+, 353 K); the lengths are those for which this study provides evidence, not universal minima. Five recommendations hold for every purpose: check that the potential energy has stopped drifting before production starts, report the lag and the estimator of any transport label together with the effective exponent of the population, prefer origin-averaged estimates to end-point ones, never read a within-run error bar as the distance to the long-time value, and remember that a lithium self-diffusivity ranks mobility, not conductivity: ion pairs and clusters that move together raise D_{app} without carrying net charge [28], so candidates selected by it should be checked for ion correlations and transference number before they are called good electrolytes.

3.9. Limitations

All systems share one force field, one temperature and one salt; the conclusions hold for three properties, the lithium and anion diffusivities and the ionic conductivity, but the numbers are specific to this setting, and the workflow, not the values, is what transfers to other campaigns. The long runs of the literature polymers were chosen by the dataset's authors; the screening systems with a replica (94) or a long run (96) are distributed as the rest of the screening set, but they are few and largely the same systems (92 in common). The reference lag of 25 ns is not a converged limit and shares data with the

truncated trajectories, so prefix-based errors at long T are optimistic. Self-diffusion is only a proxy for conductivity, and the cluster Nernst-Einstein conductivity neglects correlations between different clusters [28]; finite-size effects [29] are not corrected. The boxes are small (3.0 to 4.2 nm side, 1 to 15 chains), so each initial structure samples few chain conformations and packings; the structural part of the noise may be smaller in larger boxes, which would raise the reliability of a single run. The noise model rests on 94 replicated pairs: the allocation gains are predictions of a model that the real runs, with 92 systems, do not contradict but cannot confirm; a campaign with several replicates per candidate would test them directly. The analysis is largely exploratory: windows, thresholds, estimators and strategies were compared on the same data, and only the stopping-rule settings were chosen and evaluated on separate halves of the systems.

Table 5: Decision guide for MD screening, from the results of this study.

Purpose	What to do	Evidence in this study
Separate good from poor candidates	One 5 ns run each; rank by origin-averaged D_{app}	Spearman 0.94 among widely different polymers but 0.70 among similar screening polymers (origin-averaged estimator); not for fine ranking
Find the best candidates under a fixed budget	One run each, then new-structure replicates for candidates near the cut-off; with few rounds available, one re-screening round of a shortlist	In simulations with the measured noise, boundary-focused allocation matches four uniform replicates with 1.8 runs per candidate for the top 10% and 1.2 for the top 1% (17 and 5 rounds); a single re-screening round of a shortlist of $4k$ fixed in advance saves $\times$1.8 and $\times$3.0; real runs consistent but underpowered
Decide between two similar candidates	Several new-structure runs each	One run each: noise alone gives differences of $\times$1.65 (1 SD) and up to $\times$2.67 (95%)
Train or evaluate a model on screening labels	Split by system; average replicates where available; compare test metrics with the ceiling	Correlation with single labels capped near 0.69 (95% CI 0.56 to 0.79); a fingerprint random forest reaches 0.71, and 0.78 against the mean of two runs; improvements by better models are easier to demonstrate against averaged labels
Estimate the apparent diffusivity at a stated lag	5 to 10 ns with power-law extrapolation, one step above the cheapest setting	Typical error 1.37 from 5 ns against 2.02 for a linear fit; an empirical estimate at that lag, not of the long-time coefficient (underestimates the late MSD slope)
Obtain a diffusion coefficient or compare with experiment	Simulate until the exponent is compatible with one and D_{app} is flat across lag windows	5 of 133 runs of 50 ns meet the criterion

4. Conclusions

Short MD runs provide useful screening labels, but those labels are both time dependent and noisy. Across this collection, lithium mobility decreases during a run, together with that of the polymer, while the system is still relaxing, so a 5 ns label can overstate later-time mobility. Run-to-run noise accounts for about half of the variation across candidates, with comparable contributions from initial structure and dynamics. This noise limits prediction against single-run labels; the fingerprint model already approaches that limit, and one run is better at separating very different polymers than ranking similar candidates. Simulations with the measured noise suggest a practical allocation: run each candidate once, then replicate candidates whose selection status is uncertain. This strategy can match four uniform replicates at half the cost or less, although the real-run check is too small to confirm the gain. A single re-screening round retains much of the saving, while adaptive stopping within individual runs does not improve on fixed-length runs. The anion diffusivity and the ionic conductivity show the same pattern; the anion labels are less noisy, and ranking by lithium mobility is not ranking by conductivity, so the label to screen is a chemical choice that the noise analysis does not replace. For most systems, even runs ten times longer do not establish a long-time transport coefficient.

Data and code availability

The HTP-MD trajectories and curves are publicly available from the Toyota Research Institute platform [4]. The analysis scripts, the per-trajectory result tables and the scripts that generate every number, table and figure of this paper are available at `https://github.com/fhernandogallego/md-label-reliability` and archived on Zenodo (`https://doi.org/10.5281/zenodo.23483460`); the raw trajectories are not redistributed. The analysis scripts were written with the assistance of Claude (Anthropic); the authors reviewed and verified all code, and every reported number is regenerated by the released scripts.

Supplementary material

Supplementary Material associated with this article (data set details and validation of the trajectory processing; sensitivity of the regime census and of

the lag dependence; run-to-run reproducibility; end-point estimator; paired runs; prefix-based estimators, stopping rules and held-out evaluation; within-run slowdown; lithium environment and ion aggregation; robustness of the adaptive allocation and its check on real runs) is provided as a separate file.

CRediT authorship contribution statement

[TODO: Confirmar los papeles de cada autor.] **Francisco Hernando-Gallego:** Conceptualization, Methodology, Software, Formal analysis, Writing (original draft), Supervision. **Dimitar Vasilev-Vasilev:** Software, Data curation, Investigation, Validation, Writing (review & editing). **Diego García-Santos:** Investigation, Validation, Writing (review & editing). **Alfonso Campo-Camacho:** Validation, Writing (review & editing). **Alberto Gómez-Gómez:** Validation, Writing (review & editing).

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with writing and editing the manuscript, as none of the authors is a native English speaker. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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